

EXECUTIVE SUMMARY: Demand Forecasting & Inventory Replenishment Strategy

Prepared by: Juan Cruz Pereyra Muñoz

Date: September 2026

Subject: Predictive Inventory Modeling & Stockout Prevention

1. Objective

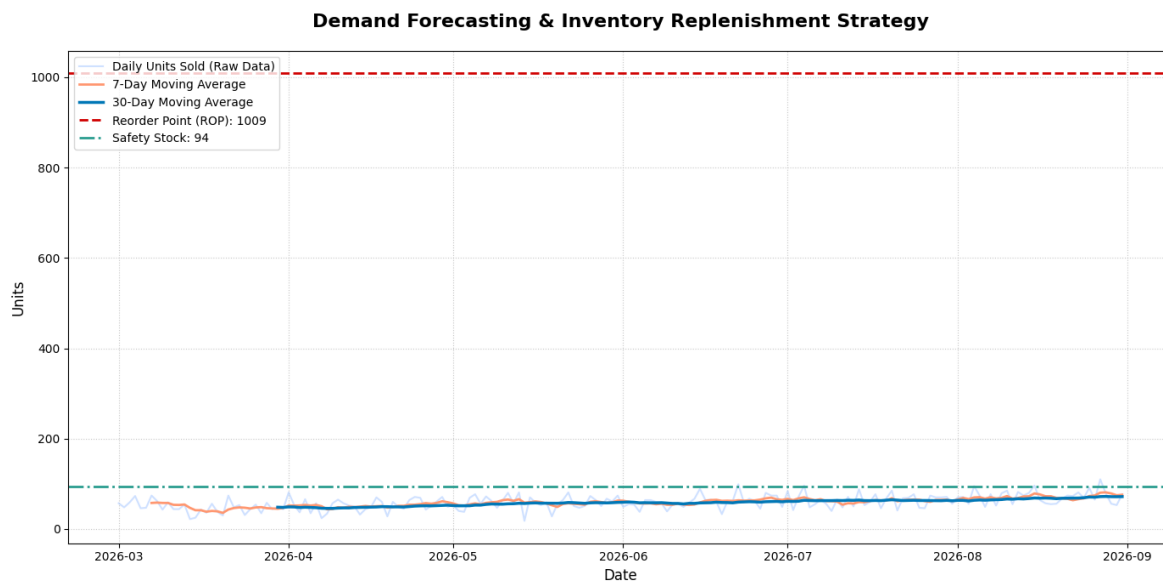
The primary objective of this analysis is to optimize inventory replenishment for a high-rotation SKU by building a predictive statistical model. By mathematically defining the Safety Stock and Reorder Point (ROP), the Operations team can prevent stockouts, protect revenue, and minimize excess inventory holding costs.

2. Methodology

A six-month historical daily sales dataset was processed using Python, NumPy, and Pandas to extract actionable forecasting metrics. The model incorporates:

- **Volatility Smoothing:** 7-Day and 30-Day Simple Moving Averages (SMA) to isolate underlying demand trends from daily noise.
- **Lead Time:** Assumed a 14-day delivery window from the supplier.
- **Service Level Target:** Set to 95% (Z-score of 1.65) to ensure high product availability.

3. Visualizing the Forecasting Model



4. Key Metrics & Actionable Triggers

Based on the standard deviation of recent demand and the target service level, the mathematical triggers for purchasing are:

- **Safety Stock Buffer: 94 units.** This is the emergency baseline required to absorb unexpected demand spikes or supplier delays without failing customer orders.
- **Strategic Reorder Point (ROP): 1009 units.** The exact threshold at which a new Purchase Order (PO) must be triggered. Waiting until inventory drops below this point guarantees a stockout before the next shipment arrives.

5. Strategic Recommendations

1. **Automate PO Generation:** Integrate the calculated ROP (1009 units) into the ERP system to automatically trigger purchasing alerts, removing human guesswork from the replenishment cycle.
2. **Dynamic Threshold Adjustments:** Recalculate the standard deviation and ROP every 30 days using this script to adapt to seasonal demand shifts.
3. **Supplier Negotiation:** Work with the procurement team to negotiate a reduction in the 14-day lead time, which would directly lower the required Safety Stock and free up working capital.